

🕒 OFF-GRID SOLAR AGRI-ENERGY MICROGRID

🔄 PM-KUSUM RETROFIT

AgroStruxure™

Solar-Synchronized Agri-Energy Microgrid & Closed-Loop Precision Irrigation

Monetizing agricultural water conservation by coupling PM-KUSUM Component B solar pump arrays with on-farm pre-cooling—turning unpumped groundwater into perishable shelf life.

41.2%

GROUNDWATER SAVED

7,000 m³/ha/yr conserved vs flood

1,696

KWH/HA FREED

Pumping electricity liberated

1.31 t/yr

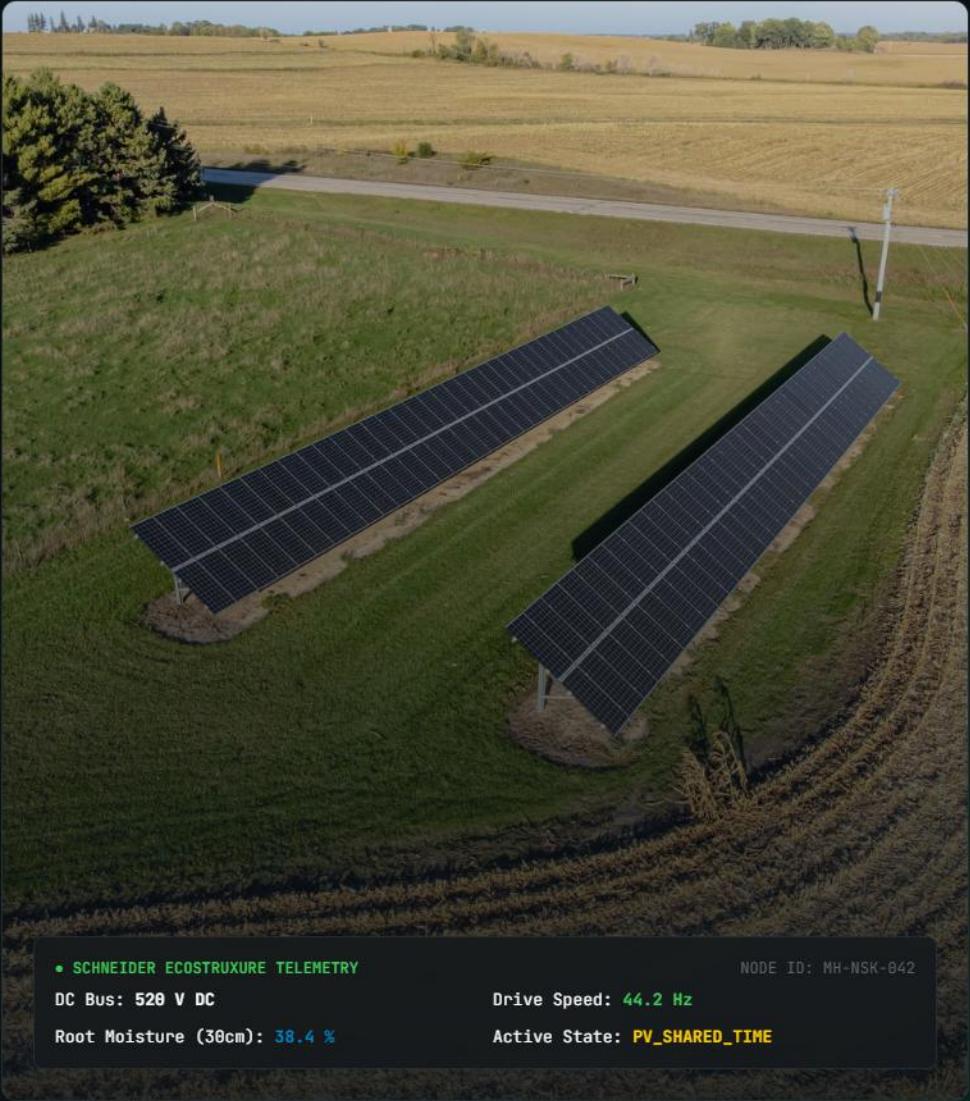
PRODUCE PRESERVED

Farm-gate heat loss eliminated

2.0 Yrs

CLUSTER PAYBACK

2 crop seasons (Tier A)



The Dual Agricultural Crisis: Aquifer Depletion & Farm-Gate Spoilage

India extracts more groundwater than China and the US combined—while ₹1.53 Lakh Crore of perishable produce rots at the farm gate.

AQUIFER EXHAUSTION

CGWB ASSESSMENT 2023

245

BCM / Year

Total Annual Groundwater Extraction

- 87% for Agriculture: Irrigation dwarfs domestic (11%) and industrial (2%) water use nationwide.
- 1,114 Assessment Units Over-Exploited: Extraction exceeds annual replenishable recharge across key breadbaskets (Punjab, Rajasthan, Maharashtra).
- 255,000 GWh Power Drain: Agricultural pumps consume 16.53% of all Indian electricity (CEA 2024).

PUMPING FLEET PROFILE

21.5M Grid Electric Pumps · 8.5M Diesel Pumps · 10.9L Solar Pumps

PERISHABLE VALUE COLLAPSE

NABCONS / MoFPI 2022

₹1.53

Lakh Crore

Annual National Post-Harvest Economic Loss

- 8.37% Farm-Gate Tomato Spoilage: Perishable losses occur immediately post-harvest due to respiration field heat (32°C+ ambient).
- Distress Selling Epidemic: Without local pre-cooling, farmers dump produce for ₹2–4/kg at APMC mandis to avoid total rot.
- The Cold Chain Paradox: Commercial cold storage sits 30–50 km away in district hubs—too far and too expensive for smallholders.

SMALLHOLDER VULNERABILITY

86.2% of Indian landholdings are < 2.0 ha (Avg: 1.08 ha)

The Solar Rebound Paradox: Why Free Power Wastes Water

PM-KUSUM Component B deployed 10.9 lakh standalone pumps, but zero marginal operating cost induces the Jevons paradox.

EMPIRICAL EVIDENCE • RAJASTHAN FIELD STUDY

Zero Marginal Cost = Unconstrained Pumping

When diesel pumps are replaced with solar arrays, operating cost drops from ₹80/hour to ₹0. Econometric research published in *Energy Policy* (Gupta, 2019) measured the real-world impact:

+16% to +39%

Increase in Groundwater Extraction After Solar Pump Adoption
Empirical measurement across groundwater-stressed Rajasthan districts (Jaipur, Sikar).

The Core Behavioral Failure: Advising a farmer to turn off their pump achieves nothing. In an off-grid setup, unconsumed solar power cannot be sold to the grid. *Stopping has zero economic value.*

KEY FINDING: Efficiency alone causes Jevons expansion; volume must be economically capped.

PHYSICAL ENERGY MISMATCH

Two-Thirds of Solar Generation Sits Idle



PLAYER / CATEGORY	CURRENT MECHANISM	THE CRITICAL FLAW
Solar Cold Rooms (Ecozen)	Standalone PV + cold room	Forces 2nd redundant PV array (₹91k wasted capex)
IoT Advisory (Fasal)	Soil moisture phone alerts	No actuation; cannot stop farmer from over-pumping
GSM Starters	Remote mobile pump on/off	Makes pumping easier; accelerates aquifer depletion

THE OPPORTUNITY: Safely divert the 63% idle midday solar surplus into farm-gate pre-cooling.

AgroStruxure™: Monetizing Water Conservation

A shared-array agri-energy microgrid that enforces a volumetric water entitlement and diverts freed solar power to cooling.

CORE INNOVATION

AgroStruxure solves the Jevons paradox by giving farmers an immediate economic asset in exchange for water not pumped: *farm-gate cold storage*.

Avoids ₹91,000 PV Capex via Time-Sharing

1. REDUCE ENERGY & WATER INTENSITY

41.2% WATER SAVED

Local FAO-56 Penman-Monteith water budgeting with dual-depth capacitive FDR probes. Cuts seasonal irrigation from 850 mm (flood) to 500 mm (pulsed drip), saving **7,000 m³/ha/yr**.

Pumping Energy Freed:

1,696 kWh / ha / year

2. MINIMIZE POST-HARVEST LOSSES

1.31 TONNES SAVED

Captures 3,187 kWh/yr (77%) of idle midday solar surplus to drive an on-farm 2 MT PCM pre-cooler at 12°C. Reduces farm-gate tomato spoilage from **8.37% down to 4.00%** (NABCONS 2022).

Preserved Produce Value:

+₹15,732 / ha / year

3. EMPOWER THE SMALLHOLDER

2.0-YR PAYBACK

Affordable industrial retrofit BOM (₹7,540 at 1,000 units). Delivers spoken vernacular audio advisories in Marathi and Hindi over WhatsApp (Krishi Mitra). 4-farm cluster net gain: **₹25,332/farm/yr**.

Cluster Capex (after 35% subsidy):

₹50,212 per smallholder

4. STRENGTHEN CLIMATE RESILIENCE

2.94 t CO₂e ABATED

Locks water extraction to CGWB/Atal Bhujal block budgets; 100% thick-edge execution on ESP32-S3 with 90-day offline flash buffer; prevents dry-run and protects deep Deccan aquifers.

Displaced Diesel & Food Waste GHG:

2.94 Tonnes CO₂e / ha / year

Closed-Loop Precision Hydraulics & Diurnal Journey

A deterministic 24-hour cycle: soft-start drip irrigation, field-capacity cutoff, and zero-flow valve latching.



DUAL-DEPTH CAPACITIVE SENSING



10cm Surface: 42% (Evap zone)
30cm Root-Zone: 38% (Active uptake)
Field Capacity: 45% (Cutoff trigger)
Wilting Point: 18% (Stress floor)

FAO-56 Dual Kc Engine: Inputs 10cm/30cm FDR probes + solar pyranometer to calculate real-time transpiration (4.8 mm/day) vs deep percolation.

Target Water Quota: 3,500 L / day / farm

ANTI-WATER HAMMER CUTOFF SEQUENCE

- T=0s Cutoff Triggered:** Daily quota (3,500 L) reached or root zone hits 45% Field Capacity.
- T+2s VFD Decel Ramp:** Gateway writes 0 Hz to Altivar Reg 8502; controlled 8s ramp prevents negative pressure surge.
- T+10s Zero-Flow Latching:** Pulse flow meter confirms Q = 0 LPM; bistable latching solenoid valve closes at zero velocity.
- T+15s 5s Dead-Band Dwell:** TeSys D break-before-make contactor switches DC bus to pre-cooler after voltage bleeds <50V.

Hydraulic Proof: Eliminates water-hammer shockwaves (dP > 15 bar), protecting emitter lines and pump impellers from burst failure.

WHATSAPP VOICE ADVISORY

- 📞 KRISHI MITRA AUDIO (MARATHI)**
"रामभाऊ, आज 3,500 लिटर पाणी दिले. ओलावा 44% झाला आहे. आता सोलर ऊर्जा 2 MT कोल्ड रूमला जोडली आहे. टोमॅटो लगेच साठवा."
- 📞 KRISHI MITRA AUDIO (HINDI)**
"रामजी! आज 3,500 लीटर पानी पूरा हुआ। नमी 44% हो गई है। सौर ऊर्जा कोल्ड रूम में चालू है। फसल तुरंत रूम में रखें।"

Zero Actuation Risk: LLM strictly reads edge telemetry JSON to generate audio; zero control authority.

Schneider EcoStruxure™ Architecture & Power Topology

Engineered break-before-make upstream DC changeover with 5-second dead-band eliminates drive stress.

ECOSTRUXURE™ 3-TIER STACK

Tier 3: Apps, Analytics & Services (EcoStruxure Cloud)

[MQTT](#) / [TLS](#)

- Agronomic Digital Twin & Sentinel-2 10m NDVI/NDWI Satellite Ingestion
- Deterministic Krishi Mitra Vernacular Advisory Copilot (Marathi/Hindi)
- PM-KUSUM RMS & Central Aquifer Depletion Monitoring Portal

Tier 2: Edge Control (EcoStruxure Edge)

[ESP32-S3](#) [FreeRTOS](#)

- AgroStruxure Industrial Gateway (240MHz, 16MB Flash, 90-Day FIFO Buffer)
- Modbus RTU Master Engine (CiA402 Profile, Registers 8501, 8502, 3201-3208)
- Deterministic 3-Layer Entitlement Engine & 5s Dead-Band Safety Sequencer

Tier 1: Connected Products & Switchgear

[MODBUS](#) / [HARDWARE](#)

- **Schneider Altivar Solar ATV320 VFD** (Embedded MPPT, Dry-Run Protection)
- **Schneider TeSys D Contactors** (2x LC1D09BD, Mech & Elec Interlocked)
- Dual FDR Soil Probes, 1" Hall-Effect Flow Meter, SHT31-D, Pyranometer

UPSTREAM DC BUS POWER TOPOLOGY



CRITICAL ELECTRICAL SAFETY PROOF:

Switching contactors downstream on a running VFD causes destructive $V = L \cdot (di/dt)$ inductive voltage transients that puncture IGBT gates and trip drive fault codes. Switching upstream on the DC bus with an engineered 5-second bleed-down deadband protects motor insulation, contactor contacts, and inverter electronics.

System Innovation & Competitive Defense

A multi-vector moat: time-shared solar PV capex, closed-loop entitlement governance, and thick-edge reliability.

Feature Dimension	Conventional Solar	Ecozen (EcoTron+EcoFrost)	Fasal / IoT Advisory	AgroStruxure™ (Ours)
Solar Asset Sizing	Pump only (63% wasted)	2 Separate PV Arrays (Pump + Cold)	N/A (Grid/Battery IoT)	1 Shared Array (Saves ₹91k)
Water Governance	Unconstrained Pumping	Unconstrained Pumping	Advisory Graphs Only	Volumetric Cap & Rebound Lock
Automated Actuation	Manual Switch	Independent Inverters	Zero Actuation	Automated TeSys DC Changeover
Offline Resilience	Manual DOL	Cloud Dependent	Cloud Dependent	100% Local FreeRTOS Execution
Vernacular Advisory	None	Basic English App	Mobile App Notifications	WhatsApp Voice in Marathi/Hindi
Payback Period	5+ Years (Diesel replace)	6–8 Years (High Capex)	Subscription Overhead	2.0 Years (2 Crop Seasons)

DEFENSIBLE ECONOMIC MOAT

Eliminating Redundant Solar Capex

Competitors like Ecozen treat irrigation and cold rooms as disconnected product silos, selling a 5 MT cold room with a separate 4 kWp PV array.

₹91,000

Capex Saved per Pre-Cooler Installation

Avoids 2.6 kWp of standalone PV panels (@ ₹35k/kWp) by time-sharing existing PM-KUSUM infrastructure.

- **Zero Land Footprint:** Requires no additional arable land for solar structures.
- **Drive Agnostic:** Retrofits onto Shakti, Kirloskar, or Lubi solar pumps.

PROTECTED BY VOLUMETRIC WATER ENTITLEMENT

Quantified Impact: The Single Source of Truth

Every figure reproduced directly from `impact_model.py` with explicit baseline tracking and transparent bounds.

41.2%

WATER

GROUNDWATER SAVED
7,000 m³ / ha / year

Baseline: 17,000 m³ (flood) → 10,000 m³ (entitlement scheduled across 2 tomato seasons).

1,696

ENERGY

KWH FREED ANNUALLY
3,187 kWh Surplus Captured

77% of idle solar surplus put to work; 951 kWh/yr residual uncaptured transparently reported.

1.31 t

FOOD

PRODUCE PRESERVED
+₹15,732 Revenue

Farm-gate tomato spoilage cut from 8.37% (NABCONS 2022) to 4.00% residual loss.

2.94 t

CARBON

CO₂E ABATED ANNUALLY
2.55t Diesel + 0.39t Food

Displaces diesel genset cooling emissions + embodied carbon in avoided tomato spoilage.

HONEST WATER ATTRIBUTION BREAKDOWN (PER SEASON)

TRANSITION STAGE	WATER APPLICATION	SAVINGS DELIVERED
Flood Baseline	8,500 m ³ / ha	0 m ³ (Baseline)
Flood → Conventional Drip	6,000 m ³ / ha	2,500 m ³ (29.4% saving via hardware)
Drip → AgroStruxure Entitlement	5,000 m ³ / ha	1,000 m ³ (11.8% saving + rebound locked)

ENGINEERING THERMAL PULL-DOWN VERIFICATION

Pull-down from 32°C to tomato-safe 12°C: $Q = m \cdot C_p \cdot \Delta T / t = (2000 \text{ kg} \times 3.7 \text{ kJ/kg}\cdot\text{K} \times 20 \text{ K}) / 14,400\text{s} = 10.28 \text{ kW}_{th}$ (3.43 kW_{el} @ COP 3.0)

1.8 MT Batch: 1.71 kW avg over 4h

Fits 4.8 kWp array

2.8 MT Batch: 3.43 kW avg over 4h

CHOSEN (Fits 4.8 kWp array)

5.8 MT Batch: 8.56 kW avg over 4h

EXCEEDS single-farm array

PCM Buffer & Biology: Phase Change Material maintains 12°C for 16h overnight without grid power. 12°C target strictly prevents chilling injury (<10°C, USDA Handbook 66).

Industrial BOM, Unit Economics & Scaling

A ₹7,540 edge controller BOM delivers a 2.0-year capital payback for a 4-farm pre-cooler cluster.

EDGE CONTROLLER BOM (1,000 SCALE)		TOTAL: ₹7,540
COMPONENT	DESCRIPTION	COST (INR)
ESP32-S3-WROOM-1	(16MB Flash, 8MB PSRAM)	₹420
Isolated MAX485	Transceiver + TVS Diode	₹180
Dual FDR Capacitive Soil Probes	(10cm & 30cm)	₹760
Sensirion SHT31-D	Temp/RH + Pyranometer	₹600
1-inch Hall-Effect Pulse Flow Meter		₹650
1-inch 12V Bistable Latching Solenoid Valve		₹650
Schneider TeSys D Contactors x2 (LC1D09BD)		₹2,300
24V SMPS + Opto-Isolated Relay Driver		₹380
SIM7600 4G LTE Cat-1 Cellular Module		₹620
IP67 Enclosure, DIN Rail, DC SPD, Cabling		₹980
Standalone controller payback: 1.5 crop seasons via avoided pump burnout and yield protection.		

TWO-TIER ASSET OWNERSHIP ECONOMICS

Tier A: Farm Pre-Cooler (4-Farm Cluster sharing 2 MT Unit)

2.0-YR PAYBACK

Gross Capex: ₹4,00,000 → Less Shared-Array Saving: -₹91,000 → Net: ₹3,09,000.
After 35% MIDH/AIF capital subsidy: **₹2,00,850 total** (₹50,212 per farm).

Annual Net Gain per Farm: **+₹25,332 / year**

(₹15,732 preserved produce + ₹12,000 market price timing gain - ₹2,400 opex).

Tier B: FPO Holding Hub (20-Farm Cooperative Hub, 5 MT Unit)

4.2-YR PAYBACK

Hub Capex: ₹12,00,000 (owns 4 kWp PV) → ₹7,80,000 post-subsidy.
Net Storage Revenue (@ ₹3.00/kg/stay): **₹1,84,375 / year**. Claims zero shared-array saving.

- Go-To-Market Channels:**
- 1. **PM-KUSUM Retrofit:** Drive-agnostic kit sold through FPOs & PACS under AIF 3% interest subvention.
 - 2. **Schneider OEM Skid:** Pre-assembled Altivar Solar + TeSys skid bundled for new solar EPC tenders.

ALIGNMENT: Fully compliant with Agriculture Infrastructure Fund (AIF) & MIDH subsidies

Implementation Roadmap & Multidisciplinary Team

From software simulation to a 25-farm field pilot in Maharashtra with Schneider Electric India Foundation synergy.

12-MONTH EXECUTION ROADMAP

PHASE 1 (OCT 2026)

Simulation & Truth

AgroSim mathematical model validated; CiA402 Modbus registers mapped; deterministic prompt guards verified.

PHASE 2 (NOV-DEC 2026)

HIL Test Bench

Hardware-in-the-loop test bench with physical Altivar ATV320 VFD, TeSys D contactors, and 5s deadband validation.

PHASE 3 (Q1 2027)

Nashik Field Pilot

25-farm cluster pilot with 2 FPOs in Nashik horticulture belt. Tomato harvesting trials and water metering validation.

PHASE 4 (Q3 2027+)

Commercial Scale

OEM pre-assembled skid for PM-KUSUM tenders; expansion to 250 farms across Maharashtra and Rajasthan.

EMBEDDED SYSTEMS

Firmware Lead

ESP32-S3 & Modbus Engine

Responsible for FreeRTOS tasks, CiA402 drive state machine, isolated RS485 communication, and circular FIFO flash storage.

POWER ELECTRONICS

Electrical Architect

Switchgear & Sizing

Responsible for upstream DC changeover design, TeSys D mechanical interlocks, 5s deadband bleed down, and thermal pull-down models.

AGRONOMY & SATELLITE

Data Scientist

Hydrology & AI RAG

Responsible for FAO-56 Penman-Monteith algorithms, Sentinel-2 NDVI/NDWI satellite validation, and deterministic prompt guards.

PRODUCT & IMPACT

Product Strategist

Smallholder UX & Policy

Responsible for rural smallholder interfaces, QuartzDS compliance, FPO cluster economic models, and Atal Bhujal policy integration.

Schneider Corporate Synergy: Ready for Pre-Placement Interviews (PPI), SE Ventures Incubation, and SEIF Climate Smart Village rollout.

YUVA YODHA 2026 FINALE CANDIDATE